DS701 Homework 7: Least Squares, Logistic Loss and Regularization

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SOLUTIONS

Problem 1: Least squares by hand, and as a projection

Practices: forming and solving the normal equations, and reading the result geometrically.

Fit the line $y = \beta_0 + \beta_1 x$ to the four points

$$(x, y) \in \{(1, 2),\ (2, 3),\ (3, 5),\ (4, 6)\}.$$

(a) Write the design matrix X (with a column of ones for the intercept) and the vector $\mathbf{y}$. Form $X^T X$ and $X^T \mathbf{y}$, and solve the normal equations $X^T X\beta = X^T \mathbf{y}$ for $\hat{\beta}_0, \hat{\beta}_1$. This produces a system of 2 equations in 2 unknowns thatyou can solve directly without matrix inversion. (Both are exact decimals — no rounding needed.)

$\hat{\beta}_0 =$ _____ $\hat{\beta}_1 =$ _____

(b) Compute the fitted values $\hat{\mathbf{y}} = X\hat{\beta}$ and the residual vector $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$. Verify $X^T \mathbf{r} = \mathbf{0}$ — i.e. that $\mathbf{r}$ is orthogonal to *both* columns of X. Then compute TSS, RSS and R^2.

$\hat{\mathbf{y}} =$ _____ $\mathbf{r} =$ _____ TSS = _____
RSS = _____ $R^2 =$ _____

(c) In one or two sentences: the normal equations express "residual $\perp$ column space of X". Explain what "$\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of X" means for these four points — which set is $\mathbf{y}$ being projected onto, and why the projection is the least-squares choice.

(d) Suppose a colleague accidentally adds the feature x a second time, so the design matrix becomes $X' = [\mathbf{1} \quad \mathbf{x} \quad \mathbf{x}]$ (4×3). Show that $X'^T X'$ is singular (a one-line argument or a determinant will do). What can you say about the least-squares coefficient vector now — is it unique? — and about the fitted values $\hat{\mathbf{y}}$? Give two different coefficient vectors that produce the same fit as in (a).

Two coefficient vectors $(\beta_0, \beta_1, \beta_2)$: _____ and _____

Solution

(a) $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$, $\mathbf{y} = (2, 3, 5, 6)^T$. $X^TX = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$ $(n,\ \sum x_i,\ \sum x_i^2)$, $X^T\mathbf{y} = \begin{bmatrix} 16 \\ 47 \end{bmatrix}$ $(\sum y_i,$ $\sum x_i y_i = 2 + 6 + 15 + 24)$. Normal equations: $4\beta_0 + 10\beta_1 = 16$ and $10\beta_0 + 30\beta_1 = 47$. Multiply the first by 2.5: $10\beta_0 + 25\beta_1 = 40$; subtract from the second: $5\beta_1 = 7 \Rightarrow \hat{\beta}_1 = 1.4$, then $\hat{\beta}_0 = (16 - 14)/4 = 0.5$. (Equivalently $(X^TX)^{-1} = \frac{1}{20}\begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix}$; $\det X^TX = 120 - 100 = 20$.) So $\hat{y} = 0.5 + 1.4x$.

(b) $\hat{\mathbf{y}} = (1.9, 3.3, 4.7, 6.1)^T$; $\mathbf{r} = (0.1, -0.3, 0.3, -0.1)^T$. $\mathbf{1}^T\mathbf{r} = 0.1 - 0.3 + 0.3 - 0.1 = 0$ and $\mathbf{x}^T\mathbf{r} = 0.1 - 0.6 + 0.9 - 0.4 = 0$, so $X^T\mathbf{r} = \mathbf{0}$. $\bar{y} = 4$: TSS $= 4 + 1 + 1 + 4 = 10$; RSS $= 0.01 + 0.09 + 0.09 + 0.01 = 0.20$; $R^2 = 1 - 0.20/10 = 0.98$ (and ESS $= 9.8$, TSS $=$ RSS $+$ ESS as it must with an intercept).

(c) The column space of X is the plane in $\mathbb{R}^4$ spanned by $\mathbf{1} = (1, 1, 1, 1)$ and $\mathbf{x} = (1, 2, 3, 4)$ — the set of all vectors of fitted values $\beta_0\mathbf{1} + \beta_1\mathbf{x}$ any line could produce. $\mathbf{y}$ does not lie in that plane (no line goes through all four points), so the closest achievable $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto it: the residual $\mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to the plane (that is exactly $X^T\mathbf{r} = \mathbf{0}$), and perpendicular means shortest, i.e. smallest $\|\mathbf{y} - X\beta\|^2$.

(d) The second and third columns of X' are equal, so rank $X' = 2 < 3$ and X'^TX' (whose second and third rows/columns are identical) has determinant 0 — singular. The normal equations still have solutions but not a unique one: only $\beta_1 + \beta_2$ is pinned down (it must equal 1.4), so there is a whole line of minimizers. The fitted values are unchanged — $\hat{\mathbf{y}}$ is the projection onto the *same* column space, and the projection is unique even when its coordinates are not. Two examples: $(\beta_0, \beta_1, \beta_2) = (0.5, 1.4, 0)$ and $(0.5, 0.4, 1.0)$ (or $(0.5, 2.4, -1.0)$), all giving $\hat{\mathbf{y}} = (1.9, 3.3, 4.7, 6.1)$. *Common mistake:* claiming the fit itself breaks — it is the coefficients, not the predictions, that become non-identifiable (and, numerically, huge and unstable when the columns are merely *nearly* equal — multicollinearity).

Problem 2: Logistic loss and one gradient step

Practices: the sigmoid, the log-loss, and the chain rule that makes its gradient so simple.

Consider a one-feature logistic model $p(y{=}1 \mid x) = \sigma(wx + b)$ with $\sigma(z) = 1/(1 + e^{-z})$. Current parameters: $w = 1$, $b = -1$. Three labelled points:

$$(x, y) \in \{(2, 1),\ (0, 0),\ (-1, 1)\}.$$

The **log-loss** (negative log-likelihood) of one example is $\ell_i = -[y_i \log p_i + (1 - y_i)\log(1 - p_i)]$ with $p_i = \sigma(wx_i + b)$; the total loss is $\ell = \sum_i \ell_i$. Minimizing ℓ is the same as maximizing the likelihood L. You may use $e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$, $\ln 0.731 \approx -0.313$, $\ln 0.119 \approx -2.127$, and $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ (take it as given, or verify it in one line).

(a) For each point compute $z_i = wx_i + b$, $p_i = \sigma(z_i)$ (3 decimals) and ℓ_i. Report the total loss ℓ. Which point is the model most wrong about? You can use a calculator to compute the sigmoid and log-loss. Problems like this will be simplified for the quiz.

x_i	y_i	z_i	p_i	ℓ_i
2	1	_____	_____	_____
0	0	_____	_____	_____
-1	1	_____	_____	_____

Total $\ell =$ _____ Most wrong about the point with $x =$ _____

(b) Show, for a single example, that $\dfrac{\partial \ell_i}{\partial w} = (p_i - y_i)\, x_i$ and $\dfrac{\partial \ell_i}{\partial b} = p_i - y_i$. (Chain rule: $\ell_i \to p_i \to z_i \to w$; the given σ' does the heavy lifting.)

(c) Using (b), compute $\partial \ell / \partial w$ and $\partial \ell / \partial b$ summed over the three points, and take one gradient-**descent** step with learning rate $\alpha = 0.5$: $w \leftarrow w - \alpha\, \partial \ell / \partial w$, $b \leftarrow b - \alpha\, \partial \ell / \partial b$. Report the new (w, b) to 3 decimals. In which direction did b move, and why does that make sense given (a)?

$\partial \ell / \partial w =$ _____ $\partial \ell / \partial b =$ _____ new $(w, b) =$ _____

(d) (Optional) A classmate proposes fitting the same model by minimizing squared error $\sum_i (p_i - y_i)^2$ instead. Give two reasons, in a short paragraph, why the lecture fits logistic regression by maximum likelihood (i.e. minimizes the log-loss) rather than squared error. Hint: what does the least-squares criterion assume about the noise, and how large a penalty does each loss assign to a confident wrong prediction such as $p_i = 0.99$ when $y_i = 0$?

Solution

(a) $x = 2$: $z = 1$, $p = 1/(1 + 0.368) = 0.731$, $y = 1$: $\ell_1 = -\ln 0.731 = 0.313$. $x = 0$: $z = -1$, $p = 1/(1 + 2.718) = 0.269$, $y = 0$: $\ell_2 = -\ln(1 - 0.269) = -\ln 0.731 = 0.313$. $x = -1$: $z = -2$, $p = 1/(1 + 7.389) = 0.119$, $y = 1$: $\ell_3 = -\ln 0.119 = 2.127$. Total $\ell = 0.313 + 0.313 + 2.127 = 2.753$. The third point: the model gives the label-1 point only a 12% chance, and its loss dominates.

(b) With $\ell_i = -y_i \log p_i - (1 - y_i) \log(1 - p_i)$: $\partial \ell_i / \partial p_i = -y_i / p_i + (1 - y_i)/(1 - p_i)$; $\partial p_i / \partial z_i = p_i(1 - p_i)$; $\partial z_i / \partial w = x_i$, $\partial z_i / \partial b = 1$. Multiply:

$$\frac{\partial \ell_i}{\partial z_i} = \Big(-\frac{y_i}{p_i} + \frac{1 - y_i}{1 - p_i} \Big) p_i(1 - p_i) = -y_i(1 - p_i) + (1 - y_i) p_i = p_i - y_i,$$

so $\partial \ell_i / \partial w = (p_i - y_i) x_i$ and $\partial \ell_i / \partial b = p_i - y_i$. (Verifying σ': $\sigma = (1 + e^{-z})^{-1}$, so $\sigma' = e^{-z}(1 + e^{-z})^{-2} = \sigma \cdot \frac{e^{-z}}{1 + e^{-z}} = \sigma(1 - \sigma)$.)

(c) $p_i - y_i$: -0.269, $+0.269$, -0.881. $\partial \ell / \partial w = (-0.269)(2) + (0.269)(0) + (-0.881)(-1) = -0.538 + 0.881 = 0.343$; $\partial \ell / \partial b = -0.269 + 0.269 - 0.881 = -0.881$. Step: $w = 1 - 0.5(0.343) = 0.829$ (0.828 with unrounded intermediates — either accepted), $b = -1 - 0.5(-0.881) = -0.560$. b went *up*: two of the three points are label 1 and the model is under-predicting them (net $\sum (p_i - y_i) < 0$), so raising the base rate lowers the loss (it drops from 2.75 to about 2.35). *Common mistake:* adding $\alpha \nabla \ell$ — that is ascent on the *loss*. Written as gradient ascent, the update adds α times the gradient of the *log-likelihood* — the same move.

(d) (i) Least squares is maximum likelihood only under Gaussian noise; a 0/1 label is a

Bernoulli outcome, and its likelihood is $\prod p_i^{y_i}(1-p_i)^{1-y_i}$ — maximizing that is exactly minimizing the log-loss, so log-loss is the principled objective and its minimizer gives calibrated probabilities. (ii) Squared error caps the penalty for a confident wrong answer at 1 ($p = 0.99, y = 0$: 0.98), while log-loss charges $-\ln 0.01 = 4.6$ and grows without bound — it refuses to let the model be confidently wrong. A third acceptable reason: with the sigmoid inside, squared error is non-convex in (w, b) with flat regions where the gradient $\propto p(1-p)$ vanishes for saturated wrong predictions, whereas the log-loss gradient $(p-y)x$ stays large exactly when the error is large — gradient descent behaves well.

Problem 3: Ridge shrinks, LASSO zeroes

Practices: what the L2 and L1 penalties actually do to a coefficient, and how to choose the penalty weight.

Throughout, $\lambda \geq 0$ is the penalty weight.

(a) **Ridge in one dimension.** For a no-intercept model $y \approx \beta x$ with data $(x, y) \in \{(1,1),\ (2,4),\ (3,7)\}$, the ridge objective is $J(\beta) = \sum_i (y_i - \beta x_i)^2 + \lambda\beta^2$. By setting $dJ/d\beta = 0$, show that $\hat{\beta}_\lambda = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$ (this is $(X^TX + \lambda I)^{-1}X^T\mathbf{y}$ with X a single column). Evaluate it for $\lambda = 0, 1, 10$ (3 decimals) and describe in a sentence what happens as λ grows. Can any finite λ make $\hat{\beta}_\lambda$ exactly 0?

$\hat{\beta}_0 =$ _____ $\hat{\beta}_1 =$ _____ $\hat{\beta}_{10} =$ _____ exactly 0? _____

(b) **LASSO in one dimension.** The building block of the LASSO is the scalar problem

$$\min_{\beta} \ \tfrac{1}{2}(\beta - z)^2 + \lambda|\beta|,$$

where z is the unpenalized (least-squares) value. Assume first $z > 0$. Show that the minimizer is $\hat{\beta} = z - \lambda$ if $z > \lambda$ and $\hat{\beta} = 0$ if $z \le \lambda$. (Argue in two cases: if the minimizer is $\beta > 0$, differentiate; if $\beta < 0$, show it cannot be a minimizer when $z > 0$; the remaining candidate is $\beta = 0$.) By symmetry the general answer is $\hat{\beta} = \text{sign}(z)\max(|z| - \lambda, 0)$ — the *soft threshold*. Evaluate it for $\lambda = 1$ and $z = 3$, $z = 0.5$, $z = -2$.

LASSO, $\lambda = 1$: $z = 3 \to$ _____ $z = 0.5 \to$ _____ $z = -2 \to$ _____

(c) The ridge version of the same scalar problem, $\tfrac{1}{2}(\beta - z)^2 + \lambda\beta^2$, has minimizer $\hat{\beta} = z/(1 + 2\lambda)$ (you need not derive it). Evaluate it for $\lambda = 1$ and the same three z, and then explain in two or three sentences, using what you found in (a)–(c), why the L1 penalty produces coefficients that are *exactly* zero while the L2 penalty only shrinks them. (Think about the slope of $|\beta|$ versus β^2 at $\beta = 0$.)

Ridge, $\lambda = 1$: $z = 3 \to$ _____ $z = 0.5 \to$ _____ $z = -2 \to$ _____

(d) **Scenario.** An analyst fits LASSO to predict house price from `sqft` (values in the thousands), `bedrooms` (1–5) and `age_years` (0–100), without standardizing, and chooses λ as the value that minimizes the *training* RSS. State the two things wrong with this procedure and what to do instead (one or two sentences each).

Solution

(a) $dJ/d\beta = -2\sum_i x_i(y_i - \beta x_i) + 2\lambda\beta = 0 \Rightarrow \beta(\sum x_i^2 + \lambda) = \sum x_i y_i$. Here $\sum x_i y_i = 1 + 8 + 21 = 30$ and $\sum x_i^2 = 14$, so $\hat{\beta}_0 = 30/14 = 2.143$, $\hat{\beta}_1 = 30/15 = 2.000$, $\hat{\beta}_{10} = 30/24 = 1.250$. The estimate is pulled monotonically toward 0 as λ grows (the denominator grows while the numerator is fixed), but $30/(14 + \lambda) > 0$ for every finite λ: ridge shrinks, never zeroes.

(b) Let $f(\beta) = \tfrac{1}{2}(\beta - z)^2 + \lambda|\beta|$, $z > 0$. *Case $\beta > 0$:* $f = \tfrac{1}{2}(\beta - z)^2 + \lambda\beta$, $f' = \beta - z + \lambda = 0 \Rightarrow \beta = z - \lambda$, valid (positive) only if $z > \lambda$; $f'' = 1 > 0$ so it is a minimum. *Case $\beta < 0$:* $f' = \beta - z - \lambda < 0$ throughout, so f is decreasing on $(-\infty, 0)$ and no negative point is a minimizer — moving toward 0 always helps. *Case $\beta = 0$:* the only remaining candidate; when $z \le \lambda$ the right-hand slope at 0 is $-z + \lambda \ge 0$ and the left-hand slope is $-z - \lambda < 0$, so f increases as you move away from 0 in either direction — 0 is the minimizer. Hence $\hat{\beta} = z - \lambda$ if $z > \lambda$, else 0; with the sign case, $\hat{\beta} = \text{sign}(z)\max(|z| - \lambda, 0)$. $\lambda = 1$: $z = 3 \to 2$; $z = 0.5 \to 0$; $z = -2 \to -1$.

(c) Ridge, $\lambda = 1$: $z = 3 \to 1$; $z = 0.5 \to 0.167$; $z = -2 \to -0.667$ — every value is scaled by $1/3$, none is zero. Reason: near $\beta = 0$ the L2 penalty $\lambda\beta^2$ is flat (derivative $2\lambda\beta \to 0$), so it exerts no force at zero and any nonzero z pulls the coefficient off zero — the L2 optimum balances two derivatives and lands at $z/(1 + 2\lambda)$, small but nonzero. The L1 penalty $\lambda|\beta|$ has a kink at 0 with slope $\pm\lambda$ on either side, so as long as the pull of the data $|z|$ is below λ, the penalty wins and $\hat{\beta} = 0$ exactly. That is why LASSO does variable selection and ridge does not.

(d) (1) *No standardization:* the penalty $\lambda\|\beta\|_1$ charges every coefficient the same price per unit, but a coefficient on `sqft` (thousands) is naturally small and one on `bedrooms` (1–5) naturally large, so LASSO would zero out `sqft` first for reasons of units, not

relevance. Fix: center $\mathbf{y}$ and standardize each column of X (zero mean, unit variance) before penalizing. (2) *Choosing λ by training RSS:* the training RSS is minimized at $\lambda = 0$ (any penalty can only raise it), so the rule always picks plain least squares and defeats the purpose. Fix: choose λ by cross-validation / held-out error, which is what regularization is trying to improve.